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Roll no.

Degree: B-Tech, Semester: 2nd (Backlog)
END-SEMESTER EXAMINATION, APRIL-MAY, 2024

Course Title: Digital Circuits & Systems
Course Code: INECC01

Duration: 03 Hours**Max. Marks: 40**

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Express $F = xy + xz$ in a product of maxterms form	4	CO1
1b	Minimize the following expression using K-Map and realize with NOR gates only. $F(A,B,C,D) = \sum(1,2,3,10,11,13,14,15) + d(0,12)$	4	CO1
1c	Represent -15 in signed binary form, 1's complement and 2's complement form. Use 8-bit representation	4	CO1
Q2	Attempt any 2 parts of the following		
2a	Derive the logic expression for a 2-bit magnitude comparator	4	CO2
2b	With a neat design procedure, explain the implementation of 4x1 multiplexer with gate level diagram	4	CO2
2c	Write a VHDL code for full adder in behavioural style modelling	4	CO2
Q 3	Attempt any 2 parts of the following		
3a	Explain the following parameters (i) Fan out (ii) Propagation delay (iii) Noise margin (iv) Figure of Merit.	4	CO3
3b	Draw the circuit diagram for a 2-input NAND gate in (i) TTL logic family (ii) CMOS logic family	4	CO3
3c	Implement $F=AB + CD$ using CMOS logic family	4	CO3
Q 4	Attempt any 2 parts of the following		
4a	Derive the characteristic equation for a JK flip-flop	4	CO4
4b	Differentiate between synchronous and asynchronous counters	4	CO4
4c	Draw logic diagram and Explain the operation of various shift registers viz, SISO, SIPO, PIPO	4	CO4
Q 5	Attempt any 2 parts of the following		
5a	Design a 110 Mealy sequence detector using D flip-flops	4	CO5
5b	Differentiate between FPGAs and CPLDs	4	CO5
5c	Write shote note on testing and testability of digital logic circuits	4	CO5